

HEALTHCONNECT CLINIC EXPERIENCE LAB

WEEK 6 — INTEGRATED WORKSTREAM & INTERFACE CONTROL MAP

Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Purpose: *To establish how the Data Analytics, Data Science, ML / ML Engineering and Generative AI workstreams exchange outputs, manage dependencies, integrate components, evidence resulting changes, and establish readiness for Week 7 testing.*

Document Control

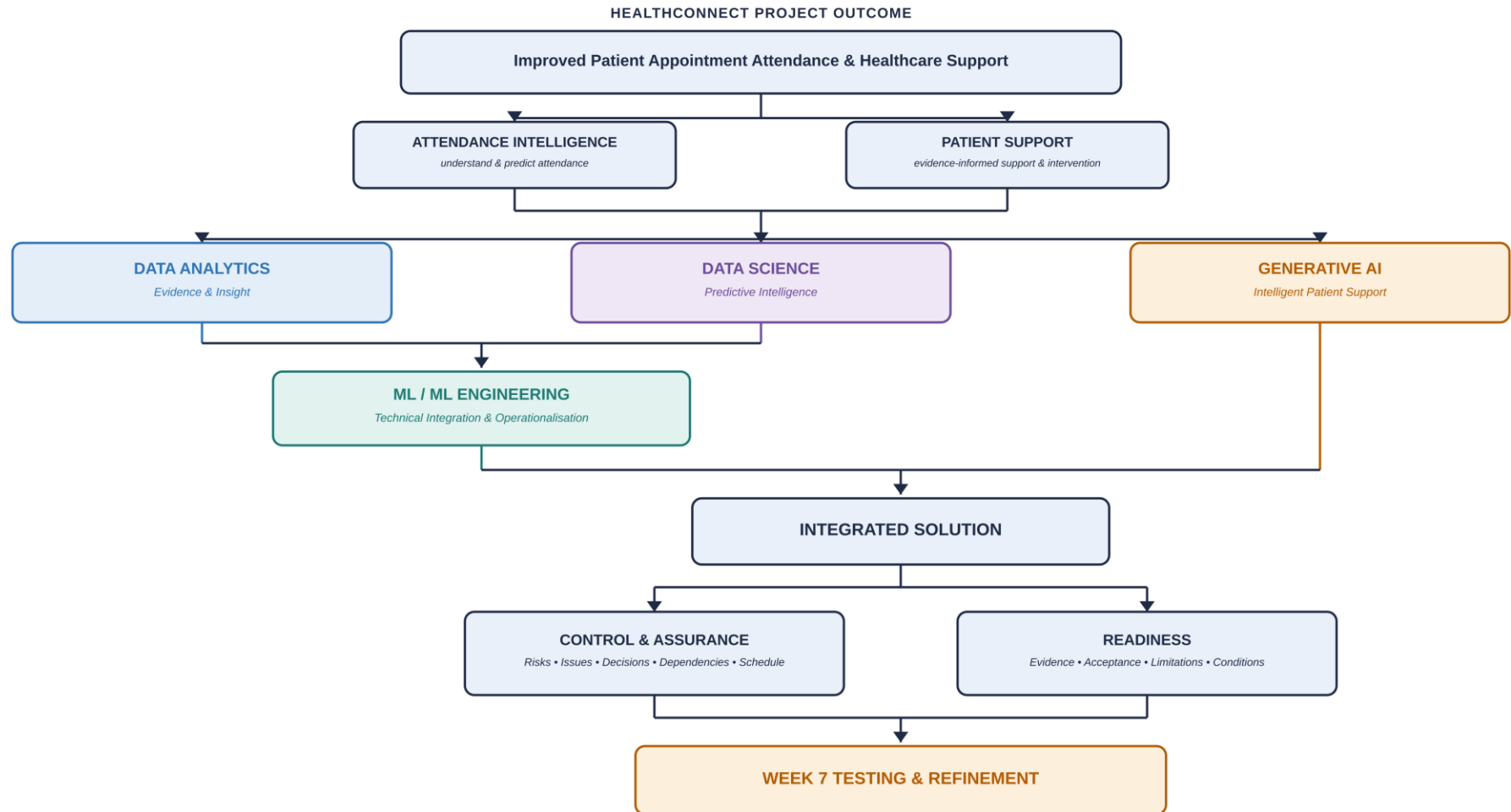
Project	HealthConnect Clinic Experience Lab
Document Type	Integrated Workstream & Interface Control Map
Reporting Period	Week 6
Tracks Covered	Data Analytics, Data Science, ML / ML Engineering, Generative AI
Document Owner	Project Management Function
Version	1.0
Status	Active

MAP STRUCTURE — 8 INTEGRATED LAYERS

Layer	Element	What It Captures
1	Project Outcome	The business outcome all four tracks collectively support
2	Workstream Outputs	What each track owns and produces
3	Input / Output Interfaces	What each track receives and provides
4	Dependency Logic	What must exist before another activity can progress
5	Integration Mechanism	How one output is actually incorporated into another workstream
6	Change / Impact	What changes because integration occurred
7	Control & Assurance	Risks, issues, decisions, acceptance and evidence
8	Testing Readiness	What integrated components move into Week 7 testing / refinement

ARCHITECTURE OVERVIEW

The integration architecture connects the project outcome, through the four specialist workstreams, into a single integrated solution, controlled release into Week 7 testing.



1. PROJECT OUTCOME LAYER

HealthConnect Desired Outcome

Improved Patient Appointment Attendance & Healthcare Support

This outcome is supported by two major capability outcomes, which give the four workstreams a common destination.



Attendance Intelligence — Understand and predict attendance behaviour.

Patient Support — Provide appropriate, evidence-informed support and interventions.

2. FOUR WORKSTREAM CAPABILITY LAYER

Each track carries an ownership identity rather than a plain label, making clear why each track exists in the integrated solution.

Data Analytics	Data Science	ML Engineering	Generative AI
Workstream	Role	Primary Contribution	
Data Analytics	Evidence & Insight	Contribution • Data-quality understanding • Attendance / no-show patterns • KPIs • Validated findings • Actionable insights • Analytical limitations	
Data Science	Predictive Intelligence	Contribution • Feature refinement • Model development • Model comparison / evaluation • Error analysis • Candidate predictive model • Model limitations	
ML Engineering	Technical Integration & Operationalisation	Contribution • Data / model interfaces • Processing pipeline • Reusable components • Integration checks • Configuration / dependency management • Reproducible implementation	
Generative AI	Intelligent Patient Support	Contribution • Support assistant capability • Knowledge-grounded responses • Prompt / workflow refinement • Safety boundaries	

Workstream	Role	Primary Contribution
		- Escalation logic - Support-use-case validation

3. INPUT / OUTPUT INTERFACE LAYER

This is the most important layer of the map. Rather than showing a plain dependency arrow (e.g. Data Analytics → Data Science), each connection is captured as a full interface contract:



The principal interface contracts identified for Week 6 are:

Provider	Output	Receiver	Purpose	Acceptance Condition
Data Analytics	Validated attendance findings	Data Science	Inform feature / model refinement	Findings sufficiently supported and interpretable
Data Analytics	Analytical requirements	ML Engineering	Inform processing / interface requirements	Requirements technically implementable
Data Science	Candidate model + feature requirements	ML Engineering	Integrate model into technical workflow	Model / interface requirements reproducible
ML Engineering	Integrated pipeline / model implementation	Data Science	Verify implementation reflects modelling requirements	Expected behaviour reproduced
Data Analytics	Evidence-based support insights	Generative AI	Inform relevant support use cases	Insight is appropriate and within approved use
Generative AI	Support capability + safety requirements	Integrated Solution	Provide patient-support functionality	Boundaries and escalation controls defined

4. DEPENDENCY LOGIC LAYER

Every major interface should answer: what is this workstream dependent on? Dependencies are classified into five types.

Dependency Type	Definition	Example
Sequential	One output must exist before another can proceed.	<i>Validated analytical findings → predictive feature refinement.</i>
Technical	A technical component must be compatible with another.	<i>Model requirements → ML pipeline implementation.</i>
Informational	One workstream requires information from another.	<i>Attendance insights → support / intervention design.</i>
Acceptance	A receiving workstream cannot treat an output as usable until a condition is met.	<i>Model output → technical integration only after interface requirements are satisfied.</i>
Governance / Safety	A component cannot progress without defined controls.	<i>GenAI capability → testing only after knowledge boundaries and escalation conditions are established.</i>

This classification gives the map genuine dependency intelligence, rather than treating every connection as equivalent.

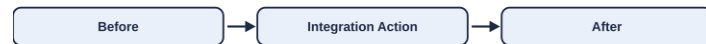
5. INTEGRATION MECHANISM LAYER

This layer distinguishes collaboration from integration by classifying the actual level of incorporation for each interface.

Mechanism	Definition
Review	Output examined but not yet incorporated.
Incorporation	Output directly used in another workstream's work.
Refinement	Receiving workstream modifies its approach because of the input.
Configuration	Technical component adjusted to accommodate another output.
Co-development	Two workstreams jointly produce a component.
Validation	One workstream evaluates another's integrated output.
Resolution	A conflict, incompatibility or dependency is identified and resolved.
Controlled Handoff	Output formally transferred with readiness conditions and evidence.

6. RESULTING CHANGE LAYER

For every meaningful integration, the map records what actually changed — giving an auditable answer to "so what actually changed?"



Example

Before: Candidate features based primarily on modelling requirements.

Integration action: Validated attendance findings incorporated into feature refinement.

After: Feature set reflects observed attendance behaviour and analytical evidence.

7. PM CONTROL & ASSURANCE LAYER

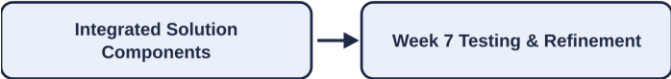
This layer runs horizontally across all four workstreams and is what makes the map unmistakably Project-Management-owned.

- **Dependencies** → what does each workstream need?
- **Handoffs** → has the output actually moved?
- **Integration** → has it been incorporated?
- **Issues** → what is preventing progress?
- **Risks** → what could compromise integration?
- **Decisions** → what requires project-level direction?
- **Schedule** → is an interface affecting sequencing or milestones?

- **Evidence** → can completion / integration be demonstrated?
- **Readiness** → is the output safe and sufficiently mature for downstream use?

8. WEEK 7 READINESS LAYER

The map does not end at "integrated." It ends at:



Each track feeds specific testable elements into Week 7:

Track	Week 7 Readiness Contribution
Data Analytics	Validated findings, KPI logic, analytical assumptions and limitations.
Data Science	Candidate model, evaluation evidence, known weaknesses and acceptance criteria.
ML Engineering	Integrated pipeline / components, interface checks, reproducibility / configuration evidence.
Generative AI	Refined assistant, test scenarios, grounding / safety / escalation criteria.

Integration Readiness Gate

Are the components sufficiently integrated, evidenced and controlled to enter testing?

Ready	Conditionally Ready	Not Ready	Blocked
-------	---------------------	-----------	---------

End of Week 6 — Integrated Workstream & Interface Control Map